

RAKSHA AI

Reasoning-based Autonomous Knowledge & Security Hardening Agent

Air-gapped autonomous cyber reasoning — find, fix, prove

AI KAVACH • CRS

EME School, Vadodara • Indian Army

34%

Best frontier-agent patch rate on SEC-bench

18%

Best frontier-agent PoC generation rate

\$152

Average cloud cost per task at DARPA AIXCC

ZERO

Bytes of source code leaving the perimeter

THE GAP

1

Deployability

Armed Forces code lives on air-gapped networks. Every existing cyber reasoning system depends on foreign cloud LLM APIs and cannot be fielded there.

2

Abandonware

All seven DARPA AIXCC systems were open-sourced, yet remain bound to competition cloud infrastructure that no longer exists.

3

Unproven noise

AI scanners flood reviewers with findings that have no reproducer. curl shut its bug bounty after AI submissions overwhelmed maintainers.

THE IDEA

1

One box, fully offline

Two 8B models at Q8 on 32 GB VRAM. No API key, no egress, no foreign dependency.

2

The verifier is the brain

The model never decides what is true. Sanitizers, taint tracking and differential testing decide; the LLM only proposes.

3

No proof, no report

Every finding ships with an executable reproducer and every patch with a five-check evidence bundle.

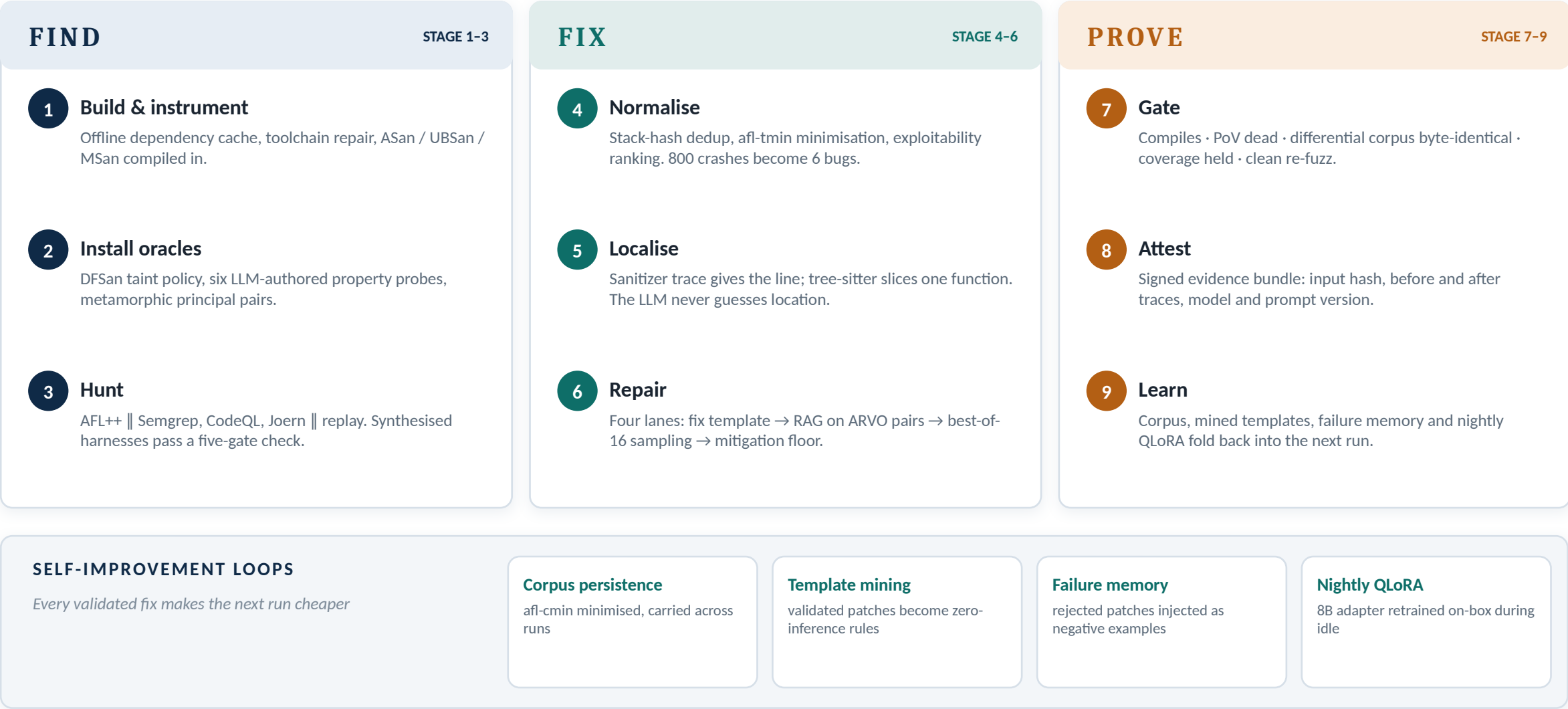
Motivation • Frontier models lose two-thirds of the time on this task. The losses are in the scaffold, not the parameter count — so a small offline model with a strict verifier is not a compromise, it is the design.

Detailed methodology

AI KAVACH

Nine stages, one pipeline. Deterministic tools do the deciding; the 8B model only proposes.

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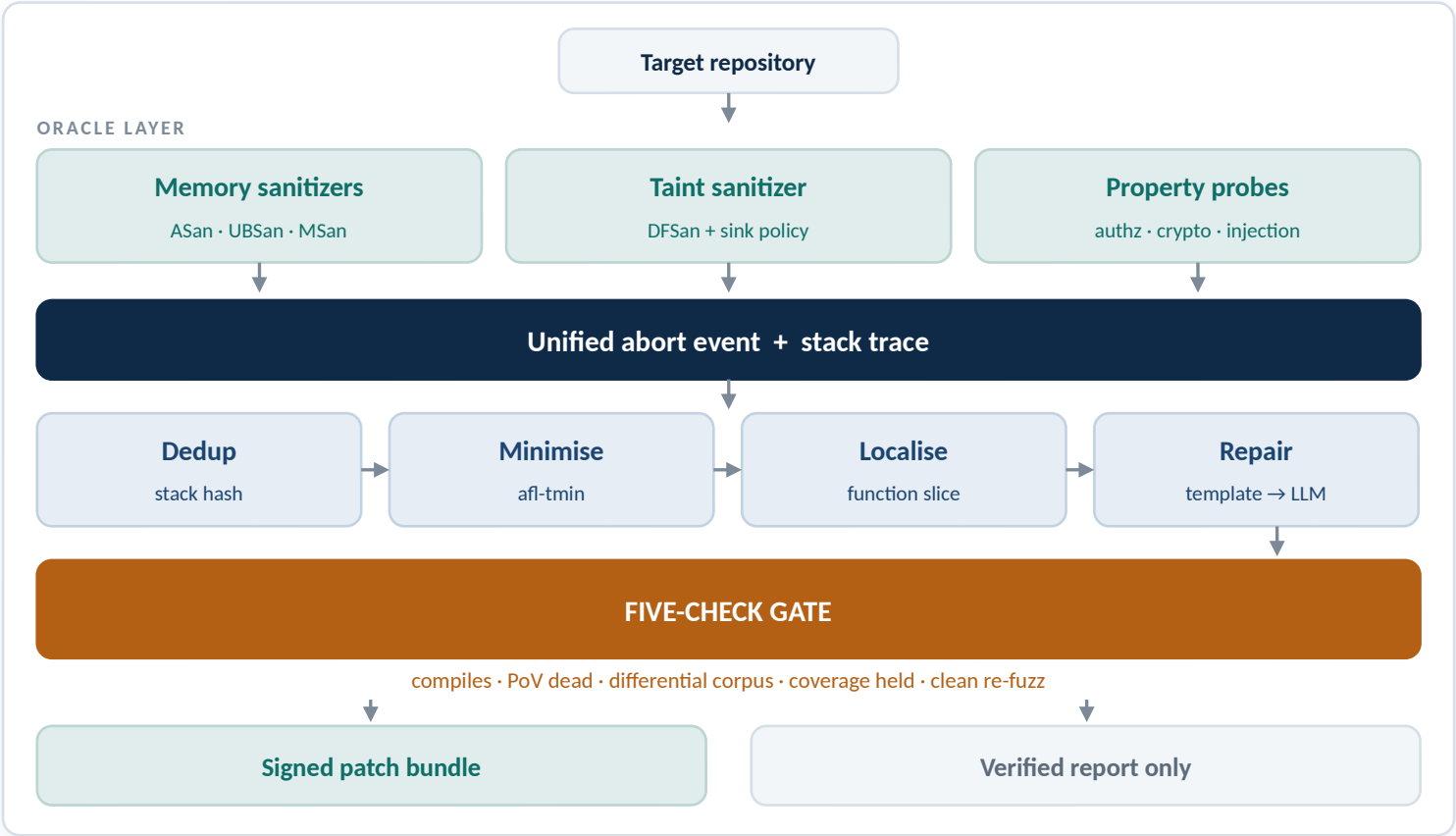


Architecture & technology stack

AI KAVACH

Pluggable oracle layer converts every bug class into one abort event, then one pipeline and one gate.

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TECHNOLOGY STACK

Inference
vLLM · Qwen3-Coder 8B Q8 · second 8B family · BGE-M3 embeddings · FAISS

Dynamic
AFL++ · libFuzzer · honggfuzz · ASan / UBSan / MSan · DFSan

Static
Sempreg · CodeQL · Joern CPG · tree-sitter AST splice

Orchestration
FastAPI · Redis queue · Docker Compose · git worktrees

Corpora
ARVO 6,100 bugs · SEC-bench · OSS-Fuzz-Gen · Fuzz Introspector

Interop
OSS-CRS manifest (OpenSSF) · LiteLLM local endpoint · ensemble fallback

EQUIPMENT	32 GB	2 × 8B	16	100 GB	0	4B CPU
	VRAM, single node	Q8, concurrent	cores for fuzzing	disk, offline cache	network interfaces	degraded profile

Salient features & novelty

USP

Runs where classified code lives — and refuses to speak without proof.

1 Oracle installation, not discovery

Non-crashing bugs get a runtime invariant injected at the sink, so authz, injection and crypto flaws abort like memory bugs — and reuse the entire pipeline instead of needing a second one.

2 Differential corpus proof

Every non-crashing fuzz input must produce byte-identical output before and after the patch. This kills the overfitting fix that silently passes a thin test suite.

3 Five-gate harness synthesis

Compiles · survives random input · reaches target · coverage still climbing · deterministic. Automates the validity check that OSS-Fuzz-Gen leaves to a human.

4 Verifier-centric small model

Frontier agents score 34% on this task. The losses are in the scaffold, so we compete there — two 8B models in 32 GB, not a cloud cluster.

5 Fail-closed autonomy

No validated patch means no patch. A verified vulnerability report is issued instead. Nothing unproven ever reaches a live military system.

6 Four compounding loops

Corpus, mined fix templates, failure memory and nightly on-box QLoRA. Run two is measurably faster than run one on identical input.

ADVANTAGE OVER EXISTING APPROACHES

AI SAST dashboard

Lists findings with no reproducer. Floods scarce cleared reviewers with unproven noise.

Cloud CRS fork

Requires egress to a foreign API. Structurally undeployable on a classified network.

RAKSHA AI

Executable proof on every output, zero egress, one node, and it gets faster each run.

Final deliverables

Everything ships as an offline bundle installable from removable media onto an air-gapped node.

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PROJECT DELIVERABLES

- **Offline deployment bundle**
One Docker Compose file, air-gap installable, no network calls at runtime.
- **RAKSHA orchestrator**
FastAPI service, oracle plugin API, four patch lanes, bandit scheduler.
- **Property probe library**
Six CWE classes across C/C++ and one scripting runtime, with policy files.
- **Fine-tuned 8B adapter**
QLoRA weights trained on validated ARVO fix pairs, shipped with the bundle.
- **Live metrics dashboard**
Precision, time-to-patch, VRAM in use, and the run-over-run learning curve.
- **Signed evidence bundles**
Per-patch provenance ledger for signals-authority sign-off before deployment.

PERFORMANCE OBJECTIVES

Report precision — findings with a reproducible PoV	100%
Patch validity — survives differential corpus test	≥ 90%
Median time to first proof-of-vulnerability	< 15 min
Median time to a fully validated patch	< 30 min
Run-2 speed-up from the four learning loops	≥ 2×
Held-out benchmark — ARVO bugs + non-crash cases	20 + 5
Peak footprint, single node, no network	32 GB

PROOF-OF-CONCEPT DEMONSTRATION

Live, autonomous, no operator input

1 Ingest

An unseen repository with no fuzz harness, on a machine with its network interface down.

2 Autonomy

Build, install oracles, synthesise a harness, find, localise, patch and prove — end to end.

3 Rigour

An overfitting patch is deliberately injected and publicly rejected, then the signed bundle is issued.